EXPLORATORY DATA ANALYSIS OF GLOBAL LAYOFFS (2020–2025)

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Table of Contents

1. Abstract

This project performs an Exploratory Data Analysis (EDA) on a global layoffs dataset covering the period from 2020 to 2025. The analysis aims to identify workforce reduction patterns across companies, industries, countries, and time periods. Various statistical and visualization techniques were used to uncover meaningful insights and understand the factors associated with layoffs.

2. Introduction

In recent years, organizations across the world have experienced significant workforce reductions due to economic uncertainty, market shifts, inflation, changing consumer behavior, and business restructuring initiatives. Layoffs affect not only companies but also employees, industries, and economies.

This project analyzes global layoff events to understand how layoffs are distributed across different industries, countries, and organizations while identifying major trends and patterns.

3. Problem Statement

Organizations across industries have experienced workforce reductions, but the extent and distribution of layoffs vary significantly. Understanding these patterns can help businesses improve workforce planning, financial management, and risk mitigation strategies.

4. Objectives

The objectives of this project are:

- Analyze layoff trends over time.
- Identify companies with the highest layoffs.
- Determine the most affected industries.
- Analyze country-wise layoff distribution.
- Examine workforce reduction percentages.
- Study the relationship between company funding and layoffs.
- Generate actionable business insights.

5. Dataset Description

Dataset Name: Global Layoffs Dataset, Source: Kaggle

Dataset Statistics:

Total Records: 3,167

Total Companies: 2,171

Industries: 30

Countries: 57

Time Period: 2020–2025

Total Employees Affected: 685,227

Dataset Features:

- Company
- Location
- Total Laid Off
- Date
- Percentage Laid Off
- Industry
- Source
- Stage
- Funds Raised
- Country
- Date Added

6. Data Cleaning & Preprocessing

The following preprocessing steps were performed:

1. Imported dataset using Pandas.
2. Checked data structure and data types.
3. Converted Date column into datetime format.
4. Created Year and Month features.
5. Checked missing values.
6. Checked duplicate records.
7. Prepared data for analysis and visualization.

Missing Values Analysis

	Missing Values	Percentage (%)
percentage_laid_off	769	24.28
total_laid_off	672	21.22
industry	1	0.03
country	1	0.03
stage	1	0.03
location	0	0.00
company	0	0.00
source	0	0.00
date	0	0.00
funds_raised	0	0.00
date_added	0	0.00

Observation:

The columns Percentage Laid Off and Total Laid Off contain missing values. Since these values represent actual layoff information, they were retained rather than imputed to avoid introducing inaccurate estimates.

Duplicate Records

```
# Checking duplicate rows
print("Duplicate Rows:", df.duplicated().sum())
```

```
Duplicate Rows: 0
```

Observation: No duplicate records were found in the dataset.

7. Exploratory Data Analysis

7.1 Dataset Overview

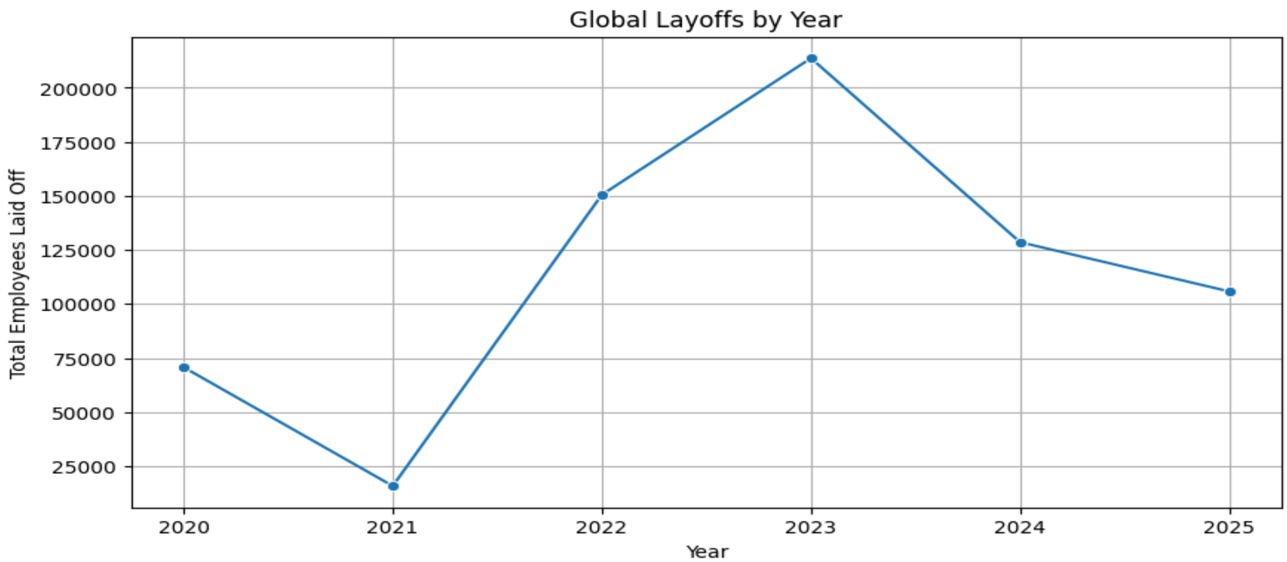
```
Total Records: 3167
Unique Companies: 2171
Unique Industries: 30
Unique Countries: 57

Date Range:
From: 2020-03-11 00:00:00
To: 2025-12-11 00:00:00

Total Employees Laid Off:
685227.0
```

Rows and Columns: (3167, 11)

Observation:

The dataset contains information from 57 countries and 30 industries, making it suitable for large-scale workforce trend analysis.

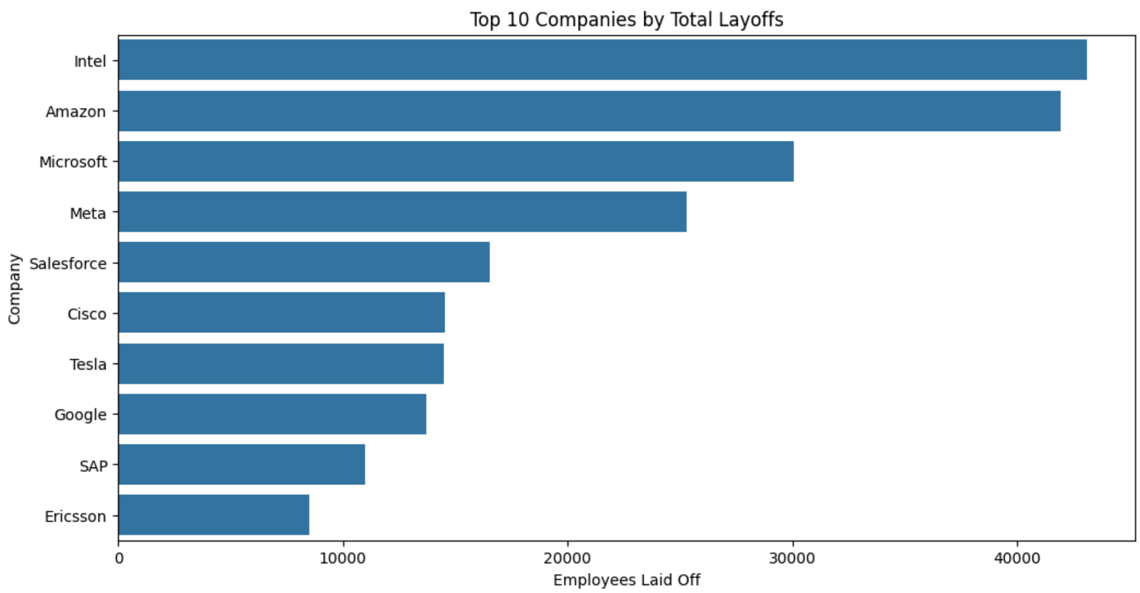
7.2 Year-wise Layoff Trend Analysis




```
Year
2020     70735.0
2021     15810.0
2022    150541.0
2023    213725.0
2024    128592.0
2025    105824.0
Name: total_laid_off, dtype: float64
```

Analysis: Layoffs increased significantly after 2021 and reached their highest level in 2023. The trend indicates large-scale workforce restructuring during this period. After 2023, layoffs began declining, suggesting a gradual stabilization of employment conditions.

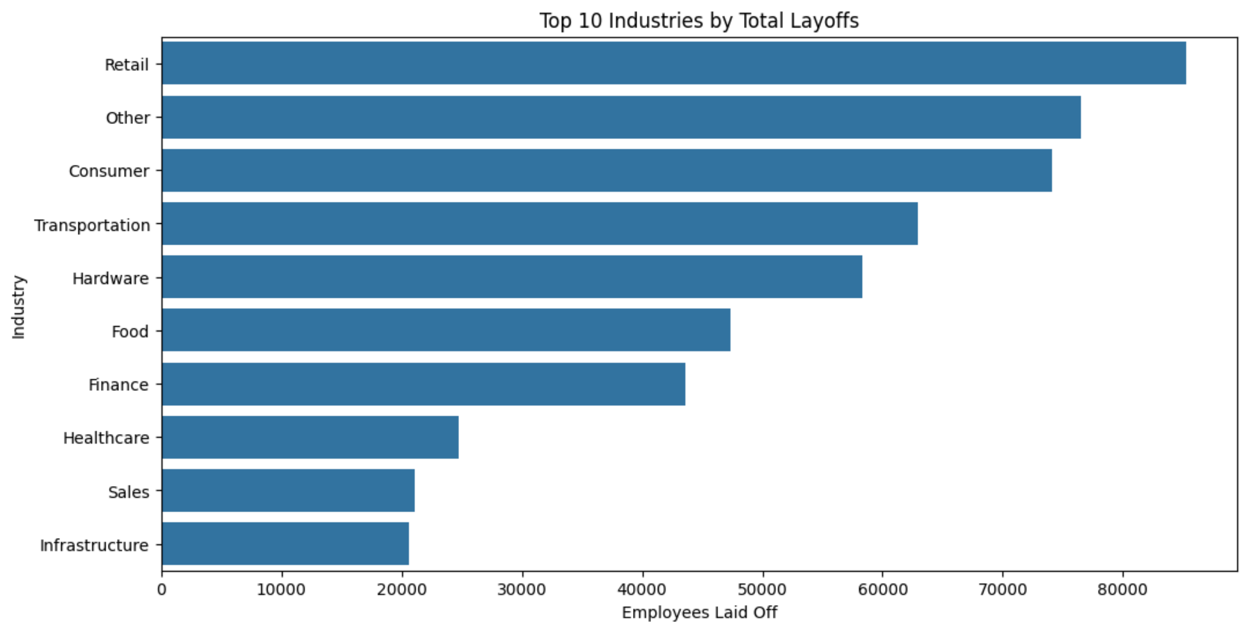
7.3 Top Companies by Layoffs




```
company
Intel         43115.0
Amazon        41940.0
Microsoft     30055.0
Meta          25300.0
Salesforce    16525.0
Cisco         14521.0
Tesla         14500.0
Google        13697.0
SAP           11000.0
Ericsson       8500.0
Name: total_laid_off, dtype: float64
```

Analysis: Technology companies dominate the list of highest layoffs. Despite being industry leaders, these organizations implemented workforce reductions to optimize costs and align with changing business priorities.

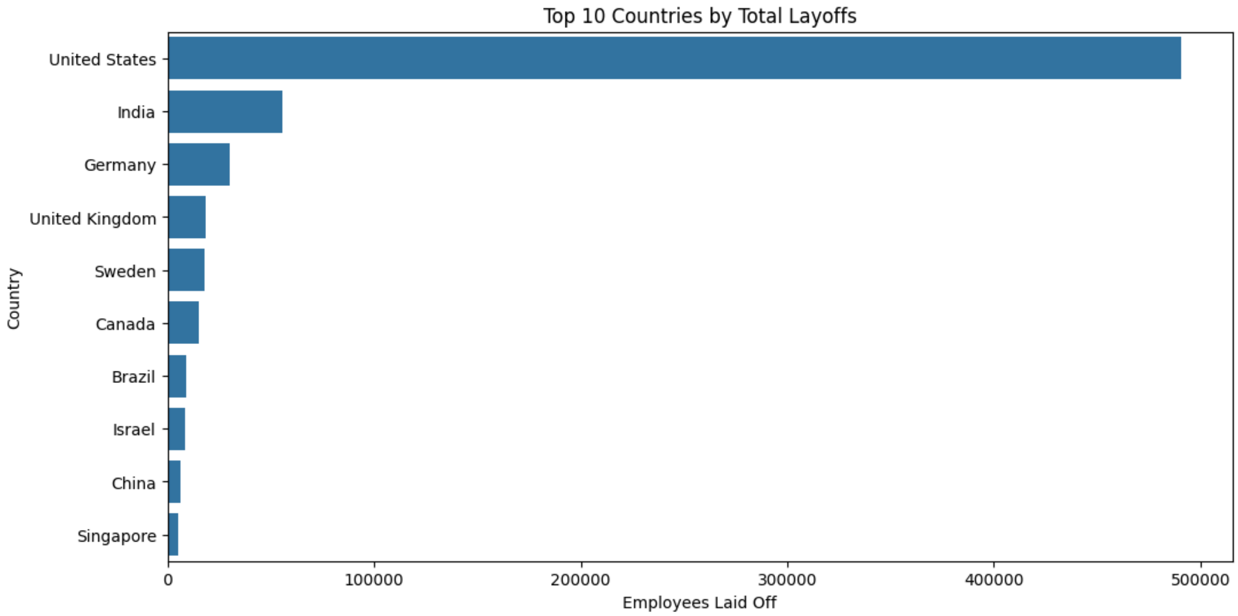
7.4 Industry-wise Layoff Analysis




```
industry
Retail            85222.0
Other             76495.0
Consumer          74105.0
Transportation    62957.0
Hardware          58363.0
Food              47351.0
Finance           43617.0
Healthcare        24701.0
Sales             21087.0
Infrastructure    20590.0
Name: total_laid_off, dtype: float64
```

Analysis: Retail experienced the highest workforce reduction, followed by Consumer and Transportation sectors. These industries are particularly sensitive to market demand fluctuations and economic conditions.

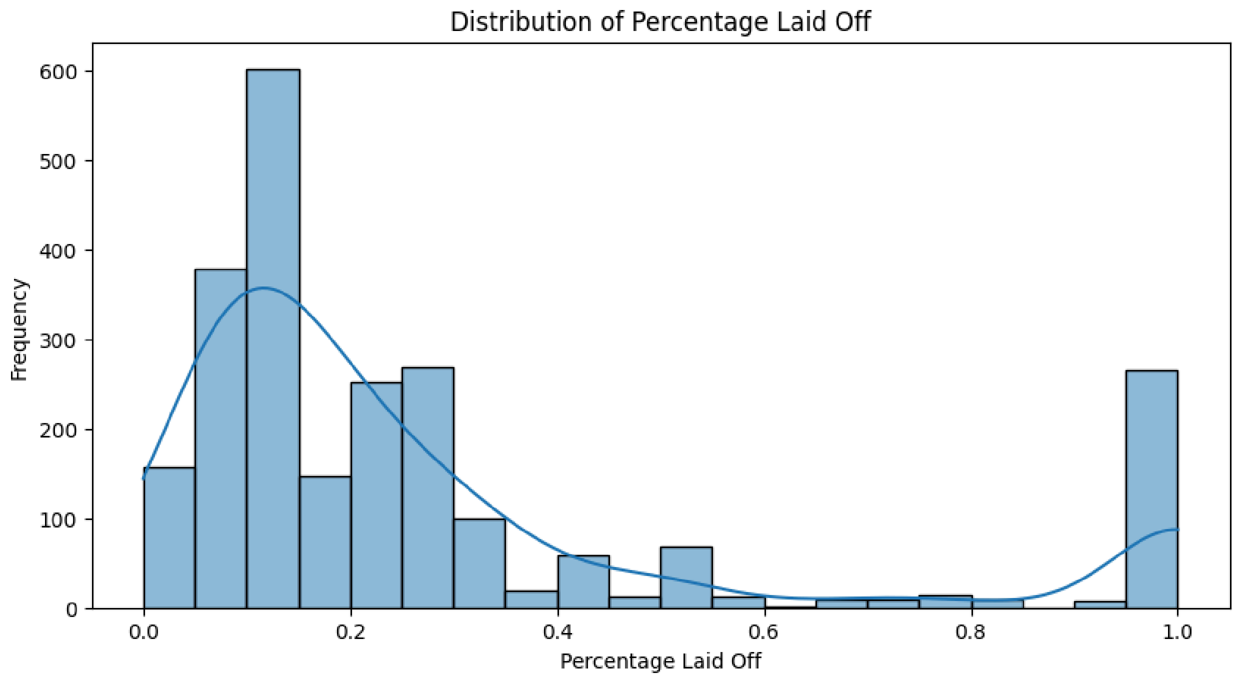
7.5 Country-wise Analysis




```
country
United States     490850.0
India              55431.0
Germany            30230.0
United Kingdom     18707.0
Sweden             17894.0
Canada             15143.0
Brazil              9026.0
Israel              8388.0
China               6205.0
Singapore           5343.0
Name: total_laid_off, dtype: float64
```

Analysis: The United States accounts for approximately 72% of all layoffs in the dataset. This reflects the country's concentration of major technology companies, startups, and multinational corporations.

7.6 Workforce Reduction Percentage Analysis




```
count    2398.000000
mean        0.280816
std         0.290316
min         0.000000
25%         0.100000
50%         0.170000
75%         0.300000
max         1.000000
Name: percentage_laid_off, dtype: float64
```

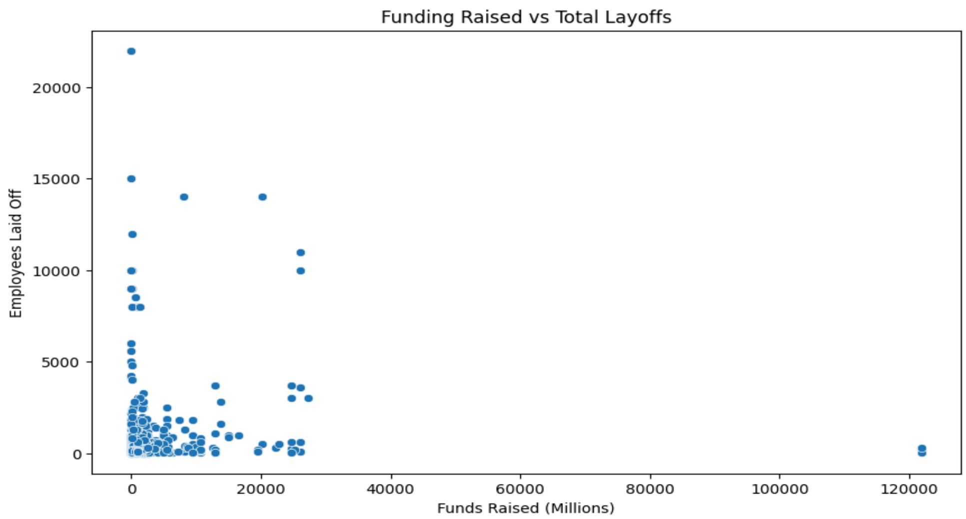
Analysis: Most organizations reduced only a portion of their workforce. However, the distribution reveals a significant number of companies that laid off all employees, indicating complete business shutdowns.

7.7 Complete Workforce Reduction Analysis

	company	industry	country
42	Advisor Credit Exchange	Finance	United States
49	Ahead	Healthcare	United States
54	Airlift	Logistics	Pakistan
71	Akudo	Finance	India
88	Alza	Finance	United States
101	Amicole	Retail	United States
107	Amplero	Marketing	United States
111	Anar	Marketing	India
119	ANS Commerce	Retail	India
130	AppHarvest	Food	United States
152	Arch Oncology	Healthcare	United States
162	Arrival	Transportation	United Kingdom
171	Assure	Finance	United States
175	Astrate Medical	Healthcare	United States
188	Atsu	Infrastructure	United States
198	Aura Financial	Finance	United States
204	Automatic	Transportation	United States
218	Awok	Retail	United Arab Emirates
243	BeepKart	Retail	India
245	Bench	Finance	Canada

Analysis: A total of 263 companies completely shut down operations. These shutdowns occurred across multiple industries and countries, highlighting the widespread impact of economic and operational challenges.

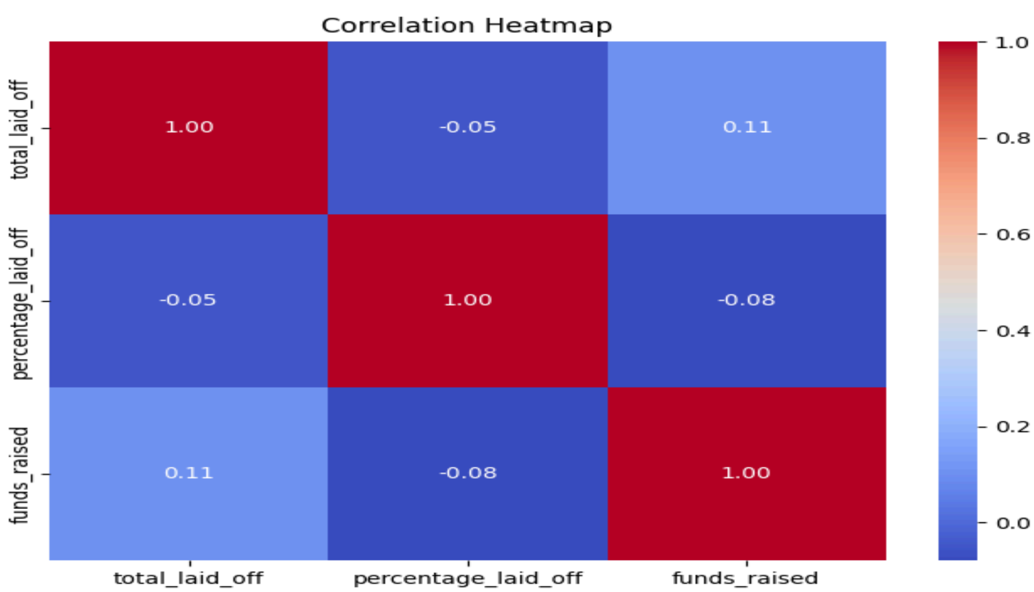
7.8 Funding Raised vs Layoffs Analysis




```
correlation = df[['funds_raised','total_laid_off']].corr()
print(correlation)

               funds_raised  total_laid_off
funds_raised       1.000000        0.105334
total_laid_off     0.105334        1.000000
```

Analysis: The scatter plot and correlation coefficient indicate a very weak relationship between funding and layoffs. Companies with significant funding also conducted layoffs, suggesting that funding alone does not determine workforce stability.

7.9 Correlation Analysis



	total_laid_off	percentage_laid_off	funds_raised
total_laid_off	1.000000	-0.047170	0.105334
percentage_laid_off	-0.047170	1.000000	-0.078746
funds_raised	0.105334	-0.078746	1.000000

Analysis: No strong relationships exist among the numerical variables. This indicates that layoffs are influenced by multiple business and economic factors rather than a single measurable variable.

8. Key Findings

• Total employees affected: 685,227

• Highest layoffs occurred in 2023

• Intel recorded the highest company-level layoffs

• Retail was the most affected industry

• United States accounted for approximately 72% of all layoffs

• 263 companies laid off their entire workforce

• Funding showed minimal correlation with layoffs

9. Recommendations

1. Improve workforce forecasting and planning.
2. Focus on sustainable growth rather than aggressive expansion.
3. Strengthen financial and operational risk management.
4. Invest in employee reskilling and upskilling programs.
5. Monitor industry trends to anticipate workforce challenges.

10. Conclusion

The analysis reveals that global layoffs between 2020 and 2025 were driven primarily by economic uncertainty and organizational restructuring. Workforce reductions affected companies of all sizes and funding levels. The findings emphasize the importance of strategic workforce planning and sustainable business growth practices.